

SUMMARY

FULL STACK DEVELOPMENT

Chapter 10

UTIL Package – Collection Framework

What is Util Package:-

The **Java.util** package is a one of a most important build-in package in a Java.

- ✓ Data storage (Collections)
- ✓ Working with dates
- ✓ Working with random numbers
- ✓ Reading input
- ✓ Handling arrays
- ✓ Using scanners, lists, maps, sets, etc.

Why do we need util.package in Java:-

Because core java does not have build-in features for many task like :

- ✓ Storing multiple object easily

- ✓ Sorting data
- ✓ Getting user input
- ✓ Working with dates and time
- ✓ Generating random numbers

Common Classes in java.util Package:-

Class Name	Purpose
Scanner	To take input from user
ArrayList	Growable array
HashMap	Key-value pair map
HashSet	Set of unique values
LinkedList	List with nodes
Collections	Utility methods (sort, reverse...)
Date	Work with dates
Random	Generate random values
Arrays	Utility methods for arrays

What is Collections Framework:-

Java Collections Framework is a set of **classes and interfaces** that help store, manage, and process groups of objects efficiently.

- ✓ Resizable data structures
- ✓ Built-in algorithms (sorting, searching, etc.)
- ✓ Easy data processing
- ✓ Provides ready-made classes like ArrayList, HashSet, HashMap

Why collections framework:-

Collections solve the limitations of arrays:

Arrays	Collections
Fixed size	Grow/shrink dynamically
Can store only same type	Can store any reference type
No built-in algorithms	Has sorting, searching, etc.
Manual operations	Ready-made utility methods

Collection framework:-

